

Name _____ Date _____ Period _____

Parent Functions IV: b^x and log base b

Algebra II Honors | Unit 1 | Day 7 | TEK A2.2.A

Objective: I can graph exponential and logarithmic parent functions, analyze their attributes, convert between exponential and logarithmic form, and explain why their domains and ranges are exchanged.

Vocabulary

Exponential function — the variable sits in the _____. **Base** — $b = 2, 10$, or _____.

Logarithm — log base b of a answers: what _____ on b gives a? So a logarithm IS an exponent.

Two forms — $b^x = a$ and log base b of $a = x$ say the _____ two ways.

Part 1 — Two Tables That Trade Places

x	-2	-1	0	1	2	3
$f(x) = 2^x$						

x	1/4	1/2	1	2	4	8
$g(x) = \log_2(x)$						

State precisely what happened between the two tables: _____

Can 2^x ever be 0 or negative? _____ So what is the domain of $\log_2$? _____

Part 2 — Converting Between Forms

Rewrite each in the other form. A logarithm is just an exponent asked for backwards.

Exponential form	Logarithmic form	Exponential form	Logarithmic form
$2^x = 8$	_____	$10^x = 1000$	_____
_____	log base 5 of 25 = 2	_____	$\log_2(1/8) = -3$
$3^x = 81$	_____	$e^x = 1$	_____

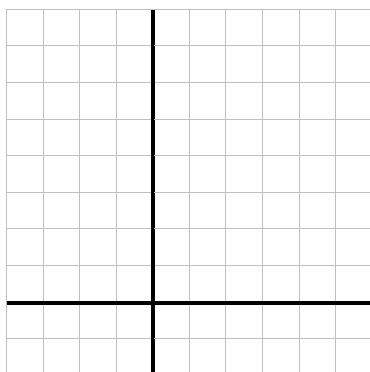
Solve by inspection: $2^x = 32 \rightarrow x =$ _____ $\log_2(64) =$ _____ $\log(100) =$ _____

Why is log base b of 1 always 0, for any base? _____

Part 3 — Graph Them Both

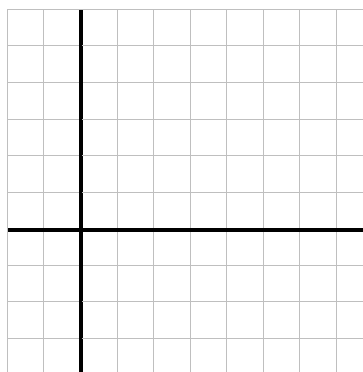
Each square is 1 unit. Draw the asymptote dashed first.

A. $f(x) = 2^x$



HA _____ intercept _____

B. $f(x) = \log_2(x)$



VA _____ intercept _____

Part 4 — Attributes Side by Side

Attribute	b^x	log base b
Domain	_____	_____
Range	_____	_____
Intercept	_____	_____
Asymptote	_____	_____

Every row is swapped. Explain what relationship between the two functions causes that:

They are _____ — formalized on Day 10.

Part 5 — Going Further

1. Which grows faster past $x = 1$, 2^x or 10^x ? Justify with two values.

2. Every exponential graph b^x passes through one shared point. Name it and explain.

3. Why is log base b of a negative number undefined?

No exponent on a positive base _____

Exit Ticket

A function has domain $(0, \infty)$, range all reals, x-intercept $(1, 0)$, and a vertical asymptote at $x = 0$.

Family: _____ Inverse family: _____ Rewrite $\log_2(16) = 4$ in exponential form: _____